

FathomNet 2025 Hierarchical Marine Species Classification Using Deep Learning Models

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Abstract

This project focuses on developing a deep learning model for the FathomNet 2025 hierarchical marine species classification challenge. The task requires the model to classify underwater marine organisms into biological categories using image data. Underwater images are challenging because they contain low lighting, haze, occlusion, complex backgrounds, and visually similar species. To address this problem, a Swin-B Transformer model was selected as the final model. Swin-B is a Vision Transformer architecture that uses shifted window attention and hierarchical feature extraction, making it suitable for fine-grained image classification. The model was trained using transfer learning with ImageNet pretrained weights, data preprocessing, augmentation, and a modified classification head for the FathomNet categories. The final system generates predictions in the required competition format, containing annotation IDs and predicted concept names. Overall, the Swin-B model showed strong potential for learning detailed underwater visual features, although it required higher computational resources compared to CNN-based models.

1. Introduction

Marine species identification is important for ocean monitoring, biodiversity research, and ecological analysis. However, manually identifying marine organisms from underwater images is time-consuming and requires expert knowledge. Deep learning can help automate this process by learning visual patterns from labelled image datasets.

The FathomNet 2025 competition focuses on hierarchical classification of marine animals. Unlike normal classification, this task considers the biological relationship between labels. For example, predicting the correct genus but wrong species is still better than predicting a completely unrelated family. Therefore, the model must not only recognize the organism visually, but also produce predictions that are close to the correct taxonomic category.

This project investigates several deep learning approaches for underwater image classification. The final selected model is Swin-B Transformer because it can capture both local details and wider image context. This is useful for underwater species recognition, where small shape differences, texture, colour, and surrounding background may affect the prediction.

2. Problem Statement

Underwater marine species classification is challenging because of several factors. First, underwater images often contain poor lighting, haze, colour distortion, and motion blur. Second, the organisms may be small, partially hidden, or surrounded by complex backgrounds. Third, some species share very similar shapes and textures, which makes fine-grained classification difficult.

Another challenge is that the FathomNet 2025 task uses hierarchical scoring. This means the model is not only judged based on exact class prediction, but also based on how close the prediction is to the correct taxonomic category. A model that predicts the correct genus but wrong species receives a smaller penalty than a model that predicts a completely different family. Therefore, the model must learn both visual features and taxonomic relationships.

The main problem addressed in this project is how to build a deep learning model that can classify marine organisms accurately while handling the difficulties of underwater imagery and hierarchical classification.

3. Project Objective

The main objective of this project is to develop a deep learning model capable of classifying marine species from underwater images using the FathomNet 2025 dataset.

The specific objectives are:

1. To preprocess and organize the FathomNet image dataset for model training.
2. To fine-tune a pretrained Swin-B Transformer model for marine species classification.
3. To apply image augmentation techniques suitable for underwater image variation.
4. To generate predictions in the required submission format.
5. To evaluate the strengths and limitations of Swin-B for hierarchical marine species classification.

4. Dataset Description

The dataset used in this project is from the FathomNet 2025 CVPR-FGVC competition. The dataset consists of underwater images containing marine organisms. The task is to classify each annotated region of interest into the correct marine category.

The dataset is provided in COCO-style JSON format. The training file contains image information, annotation information, and category labels. The test file contains the images that must be predicted for submission. Each image may contain one or more annotations, and each annotation is linked to a category.

The competition uses 79 marine animal categories. These categories may exist at different taxonomic ranks, such as family, genus, or species. This makes the task different from normal flat classification because the biological hierarchy is part of the evaluation.

For training, the region-of-interest images were cropped and arranged into folders based on their category IDs. This allowed the dataset to be loaded using PyTorch's ImageFolder format. For testing, each image was passed through the trained model, and the predicted class was mapped back to its corresponding scientific concept name.

5. Evaluation Metric

The evaluation metric used in this project is hierarchical distance. In a normal classification task, a prediction is either correct or incorrect. However, in hierarchical classification, the distance between the predicted label and the true label is considered.

For example, if the model predicts the correct genus but not the exact species, the prediction is still biologically close. This receives a smaller penalty compared to predicting a completely unrelated organism. The best possible score is 0, which means the prediction exactly matches the ground truth. The worst possible score is 12, which means the prediction is very far from the correct taxonomic label.

Therefore, a lower score is better. The final best model in this project, Swin-B, achieved the best public score of 2.53.

6. Overall System Methodology

The complete project pipeline consists of five main stages:

6.1 Data Download and ROI Cropping

The first stage involved downloading the original images using the URLs provided in the dataset metadata. Since the dataset follows COCO-style annotation format, each image contains annotation information such as bounding boxes and category IDs.

The region of interest, or ROI, was cropped from the original image using the bounding box annotation. This step is important because it removes unnecessary background and allows the model to focus more on the marine organism.

6.2 Dataset Sorting

After the ROI images were created, they were sorted into folders according to their category IDs. This folder-based structure allowed the use of PyTorch's ImageFolder dataset loader. Each folder represented one class, and each image inside the folder became a training sample for that class.

6.3 Data Preprocessing and Augmentation

The images were resized and normalized before being passed into the models. Data augmentation was also applied to improve generalization. The augmentation techniques included horizontal flipping, vertical flipping, rotation, brightness adjustment, contrast adjustment, and colour variation.

These augmentations are useful for underwater images because marine organisms may appear in different orientations and lighting conditions. Augmentation also helps reduce overfitting, especially when some categories have fewer examples.

6.4 Model Training

Several models were trained and tested. Most models used transfer learning with pretrained ImageNet weights. The original classification head of each model was replaced with a new classification layer that matched the number of FathomNet categories.

The training process used optimizers such as AdamW, loss functions such as weighted cross-entropy, and learning rate schedulers such as OneCycleLR or cosine annealing. Class weighting was used to reduce the effect of class imbalance, while label smoothing was used to prevent the model from becoming too confident.

6.5 Inference and Submission

After training, the best model was used for inference on the test set. For each test image, the model predicted a class index. This index was mapped back to the category ID and then converted into the scientific concept name.

The final submission file contained two columns:

Column	Description
annotation_id	The ID of the test annotation
concept_name	The predicted marine category name

This CSV file was then submitted to Kaggle for scoring.

7. Models Implemented

7.1 ResNet-50

ResNet-50 was used as the first strong baseline model. ResNet stands for Residual Network. It is a convolutional neural network that uses skip connections to solve the vanishing gradient problem. These skip connections allow the network to train deeper layers more effectively.

ResNet-50 is suitable as a baseline because it is stable, efficient, and widely used in image classification tasks. For the FathomNet dataset, ResNet-50 was useful because it could extract strong visual features from underwater ROI images. It was also faster to train compared to larger Transformer models.

In this project, ResNet-50 used transfer learning from ImageNet pretrained weights. The final fully connected layer was replaced with a custom classification head for the FathomNet categories. Weighted cross-entropy was applied to handle class imbalance, and label smoothing was used to improve generalization.

The ResNet-50 baseline achieved a public score of 3.45. This result showed that ResNet-50 was a reliable model, but it was not the best model overall. Its main limitation was that CNN-based local feature extraction may not capture long-range contextual relationships as effectively as attention-based models.

Strengths of ResNet-50

1. Stable and easy to train.
2. Efficient in terms of memory and inference speed.
3. Good baseline for comparison.
4. Works well with transfer learning.

Limitations of ResNet-50

1. Limited ability to capture global context.
2. May struggle with fine-grained differences between similar species.
3. The performance ceiling is lower than Swin-B.

7.2 ConvNeXt-Base

ConvNeXt-Base was tested as a modernized CNN model. ConvNeXt is inspired by Vision Transformer design principles but still uses convolutional layers. It improves the traditional CNN structure using larger kernels, depthwise convolution, GELU activation, and modern training strategies.

The reason for testing ConvNeXt was to explore whether a modern CNN could perform better than ResNet-50 while still maintaining the efficiency of

convolutional networks. ConvNeXt has strong potential because it combines the advantages of CNNs and some ideas from Transformers.

In this project, ConvNeXt-Base used ImageNet pretrained weights. The final classification layer was modified to match the FathomNet categories. Data augmentation, weighted cross-entropy, label smoothing, AdamW optimizer, and cosine annealing were used during training.

ConvNeXt showed promise, especially because it is strong at local feature extraction. However, it was more sensitive to hyperparameters and convergence stability. Within the project timeline, it did not become the best-performing model. Therefore, it was kept as an experimental model rather than the final selected model.

Strengths of ConvNeXt

1. Modern CNN architecture.
2. Strong local feature extraction.
3. More efficient than many Transformer models.
4. Suitable for image classification tasks.

Limitations of ConvNeXt

1. Requires careful hyperparameter tuning.
2. Training stability can be sensitive.
3. Did not outperform the final Swin-B model in this project.

7.3 YOLOv8

YOLOv8 was explored as a detection and classification approach. YOLO stands for “You Only Look Once.” It is commonly used for real-time object detection because it can locate and classify objects in a single forward pass.

In this project, YOLOv8 was useful for exploring localization and ROI-based classification. Since underwater images may contain complex backgrounds, isolating the marine organism can help reduce noise. The YOLOv8 experiments tested both detection-based and classification-based strategies.

Several YOLOv8 experiments were conducted:

Experiment	Model	Strategy	Public Score
1	YOLOv8m Detection	Single-stage detection baseline	7.26
2	YOLOv8m-cls	Two-stage ROI classification	3.77
3	YOLOv8l-cls	High-resolution 448px classification	3.82
4	Medium + Large Ensemble	Weighted averaging	3.60

The YOLOv8 baseline detection model achieved a score of 7.26, which showed that direct detection was not enough for this classification task. The classification version, YOLOv8m-cls, improved the score to 3.77. The ensemble method improved it further to 3.60.

Although YOLOv8 did not achieve the best final score, it was still important because it showed how localization and ROI extraction could improve classification. It also demonstrated practical hardware optimization, such as reducing batch size for high-resolution images and using automatic mixed precision to reduce GPU memory usage.

Strengths of YOLOv8

1. Very fast and suitable for real-time detection.
2. Useful for localizing marine organisms.
3. Works well with ROI extraction.
4. Easier to deploy for real-time applications.

Limitations of YOLOv8

1. More suitable for detection than fine-grained classification.
2. Lower performance compared to Swin-B.
3. High-resolution training requires careful GPU memory management.

7.4 ViT-B/16

ViT-B/16, or Vision Transformer Base with 16×16 patches, was explored as a standard Vision Transformer model. Vision Transformers divide an image into patches and process those patches using self-attention. This allows the model to learn relationships between different regions of an image.

The purpose of testing ViT-B/16 was to understand whether a pure Transformer model could perform well on underwater species classification. Compared to

CNNs, Vision Transformers can capture global context more directly. This is useful when the overall shape and surrounding image regions help identify a species.

The ViT-B/16 experiment achieved a public score of 2.92. This result was better than the ResNet-50 baseline, showing that attention-based models are promising for this task. However, ViT-B/16 was still not the best model. A limitation of standard ViT is that it uses global attention, which can be computationally expensive and may require more data to perform well.

Strengths of ViT-B/16

1. Captures global relationships in the image.
2. Performs better than the ResNet-50 baseline in this project.
3. Useful for testing Transformer-based classification.

Limitations of ViT-B/16

1. Requires more data and computation.
2. Less efficient than Swin-B for image hierarchy.
3. Did not achieve the best final score.

7.5 Swin Transformer-Base

Swin Transformer-Base, or Swin-B, was the final and best-performing model in this project. Swin Transformer is a hierarchical Vision Transformer that uses shifted-window self-attention. Instead of applying attention across the entire image at once, Swin divides the image into smaller windows and calculates attention

inside each window. Then, the windows are shifted in the next layer so that information can be shared across different regions.

This design makes Swin-B more efficient than standard Vision Transformers while still allowing the model to learn both local and global features. This is especially useful for marine species classification because organisms may have small visual differences, and their surrounding context can also provide useful clues.

Swin-B was selected as the final model because it achieved the best public score of 2.53. It performed better than ResNet-50, ViT-B/16, and YOLOv8-based approaches. The model used ImageNet pretrained weights, a custom classification head, weighted cross-entropy loss, label smoothing, AdamW optimizer, and OneCycleLR scheduler.

The final Swin-B model was trained for 15 epochs. Class weighting was applied by calculating the inverse frequency of each class. This helped the model give more attention to rare categories. Label smoothing with a value of 0.1 was used to reduce overconfidence and improve generalization.

Strengths of Swin-B

1. Best-performing model in this project.
2. Strong at fine-grained classification.
3. Captures local details and global context.
4. Hierarchical feature extraction matches the nature of the competition.
5. More efficient than standard ViT due to shifted-window attention.

Limitations of Swin-B

- 1. Requires more GPU memory than CNN models.
- 2. Slower training compared to ResNet-50.
- 3. Can overfit if regularization is not applied.
- 4. Requires careful tuning of learning rate, batch size, and augmentation.

8. Experimental Results

The following table summarizes the main model results obtained during the project.

Model / Iteration	Architecture	Key Technique	Public Score	Status
ResNet-50 Baseline	CNN	Weighted loss and label smoothing	3.45	Baseline
ViT-B/16	Vision Transformer	Standard Transformer attention	2.92	Experimental
Swin-B Default	Hierarchical Transformer	224×224 input	2.68	Improved
Swin-B Final	Hierarchical Transformer	Dynamic weights + label smoothing	2.53	Final Best Model
YOLOv8m Detection	Detection Model	Single-stage detection	7.26	Detection baseline

YOLOv8m-cls	Classification Model	ROI-based classification	3.77	Improved YOLO approach
YOLOv8l-cls	Classification Model	High-resolution 448px input	3.82	High-resolution test
YOLO Ensemble	Classification Ensemble	Weighted averaging	3.60	Best YOLO result

Based on the results, Swin-B achieved the best score of 2.53. Since the evaluation metric is hierarchical distance, a lower score indicates better performance. This means Swin-B produced predictions that were closest to the correct taxonomic labels.

The results also show that Transformer-based models performed better than the CNN baseline. ViT-B/16 improved over ResNet-50, but Swin-B improved further because of its hierarchical shifted-window design. YOLOv8 was useful for localization and ROI-based experiments, but it did not outperform Swin-B for final classification.

9. Discussion

The project results show that model architecture has a major effect on performance in hierarchical marine species classification. ResNet-50 was a strong baseline because it was stable, efficient, and easy to train. However, its convolutional structure mainly focuses on local features, which may limit its ability to capture broader image relationships.

ConvNeXt improved the traditional CNN approach by using modern design techniques, but it required more careful hyperparameter tuning. Although it had potential, it was not selected as the final model due to training sensitivity and limited time for optimization.

YOLOv8 was useful for object localization and real-time detection. The YOLO experiments showed that cropping the region of interest can reduce background noise and improve classification. However, YOLOv8 is mainly designed for detection, and its classification performance was lower than Swin-B.

ViT-B/16 showed that Transformer-based models can perform well for this task. Its public score was better than the ResNet-50 baseline, proving that attention mechanisms are useful in underwater image classification. However, standard ViT is less efficient because it applies global attention to all image patches.

Swin-B produced the best performance because it combines the benefits of Transformers with a more efficient hierarchical structure. Its shifted-window attention allows the model to focus on local regions while still sharing information across the whole image. This is very suitable for fine-grained marine classification, where the model must recognize small visual differences between similar organisms.

Another important factor was class imbalance handling. Biological datasets often contain long-tailed distributions, where some categories have many examples and others have fewer examples. Weighted cross-entropy helped the model pay more attention to rare classes. Label smoothing also helped by preventing the model from becoming too confident, which improves generalization on unseen test images.

10. Limitations

Although Swin-B achieved the best result, the project still has several limitations.

First, Swin-B requires high GPU memory. Training Swin-B is more expensive than training ResNet-50 or other smaller CNN models. This can make experimentation slower, especially on limited hardware.

Second, the model was trained mainly as a flat classification model, even though the competition uses hierarchical scoring. The loss function predicts one class at a time, but it does not directly optimize taxonomic distance. A taxonomy-aware loss function could improve performance further.

Third, underwater images are affected by lighting, blur, haze, and domain shift. Even with augmentation, the model may still struggle when test images are very different from training images.

Fourth, some categories may have fewer or more difficult examples. This can affect model performance, especially for rare or visually similar species.

11. Conclusion

This project successfully developed and compared several deep learning models for the FathomNet 2025 hierarchical marine species classification challenge. The models tested included ResNet-50, ConvNeXt-Base, YOLOv8, ViT-B/16, and Swin Transformer-Base.

ResNet-50 provided a stable and efficient baseline. ConvNeXt offered a modern CNN approach but required more careful tuning. YOLOv8 was useful for localization and ROI-based experiments, but it was not the best for final classification. ViT-B/16 showed that Transformer-based models are effective for this task. However, the best overall model was Swin-B.

Swin-B achieved the best public score of 2.53 and was selected as the final model. Its shifted-window attention, hierarchical feature extraction, and ability to capture fine-grained visual details made it suitable for underwater marine species classification. The final model was improved using transfer learning, weighted cross-entropy, label smoothing, AdamW optimization, and learning rate scheduling.

Overall, this project shows that Swin-B is a strong model for hierarchical marine species classification. It performs well because it can capture both local organism features and wider image context, which are important for distinguishing visually similar marine species.

